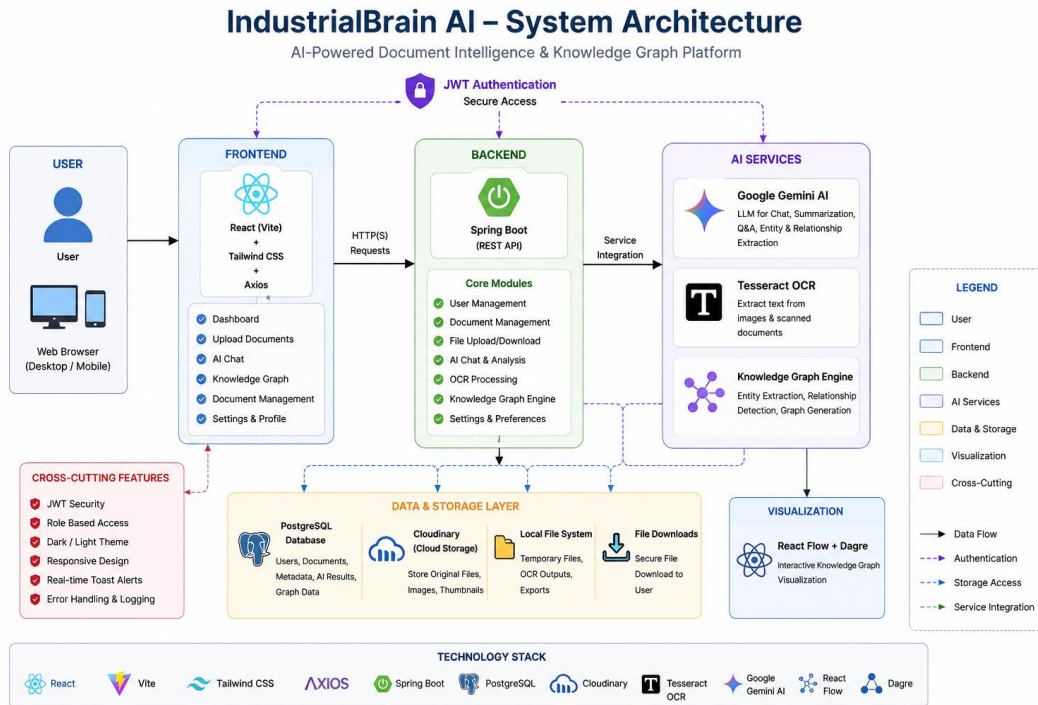


IndustrialBrain AI

AI-Powered Industrial Knowledge Platform



ET AI Hackathon 2026

AI Innovation Challenge
Project Documentation

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Submission Year

2026

Table of Contents

1. Executive Summary.....	3
2. Project Highlights.....	4
3. Problem Statement.....	5
4. Proposed Solution.....	6
5. Key Features.....	7
6. Technology Stack.....	8
7. System Architecture.....	9
8. Application Workflow.....	10
9. AI Processing Pipeline.....	11
10. Database Design.....	12
11. ER Diagram.....	13
12. Implementation Details.....	14
13. Project Screenshots.....	17
14. Security Features.....	29
15. Challenges & Solutions.....	30
16. Future Enhancements.....	31
17. Conclusion.....	32
18. References.....	33
19. Repository Structure.....	34
20. Acknowledgements.....	35

1. Executive Summary

IndustrialBrain AI is an AI-powered platform designed to transform industrial documents into an intelligent knowledge ecosystem. Organizations often manage thousands of technical manuals, Standard Operating Procedures (SOPs), maintenance reports, inspection reports, and engineering documents stored in PDF format. Retrieving critical information from these documents is time-consuming, inefficient, and largely manual.

IndustrialBrain AI addresses this challenge by combining Optical Character Recognition (OCR), Google Gemini AI, Knowledge Graph technology, and enterprise analytics into a unified platform. Users can securely upload industrial documents, extract machine-readable text, ask context-aware questions, and visualize relationships between entities through an interactive Knowledge Graph.

Built using **React, Spring Boot, PostgreSQL (Neon), Neo4j, Google Gemini AI, Tess4J OCR**, and **Cloudinary**, the platform provides secure authentication, intelligent document analysis, AI-powered assistance, and scalable knowledge management. Its modular architecture ensures maintainability, scalability, and readiness for enterprise deployment.

2. Project Highlights

- AI-Powered Industrial Document Intelligence
- Secure JWT Authentication & Authorization
- OCR using Tess4J for Scanned Documents
- Google Gemini AI Assistant
- AI-powered Contextual Question Answering
- Interactive Knowledge Graph Visualization
- Intelligent Entity Extraction & Relationship Mapping
- Dashboard Analytics & Document Insights
- Secure Cloud Storage using Cloudinary
- Modern React + Spring Boot Architecture
- PostgreSQL (Neon) Database with Neo4j Integration
- Enterprise-Ready Scalable Design

3. Problem Statement

Industrial organizations generate large volumes of technical manuals, Standard Operating Procedures (SOPs), maintenance reports, inspection documents, and engineering drawings. Most of this valuable information is stored in unstructured PDF files or scanned documents, making knowledge retrieval slow, inefficient, and highly dependent on manual searching.

Traditional document management systems primarily focus on storage rather than understanding document content. As a result, information remains scattered across multiple files, relationships between entities are difficult to identify, and engineers spend significant time searching for relevant information instead of solving operational problems.

IndustrialBrain AI addresses these challenges by integrating OCR, Generative AI, Knowledge Graph technology, and enterprise analytics into a single intelligent platform. The system transforms static industrial documents into a searchable knowledge ecosystem that improves information accessibility, accelerates decision-making, and enhances operational productivity.

4. Proposed Solution

IndustrialBrain AI provides a centralized AI-powered platform for intelligent industrial document management by integrating OCR, Google Gemini AI, Knowledge Graph visualization, and secure cloud storage into a unified enterprise solution.

The platform enables users to securely upload industrial PDF documents, extract text using Tess4J OCR, analyze document content with Google Gemini AI, and generate interactive Knowledge Graphs that reveal meaningful relationships between entities. Dashboard analytics provide insights into uploaded documents, while secure JWT authentication ensures protected access to enterprise data.

Objectives

- Digitize industrial documents using OCR.
- Enable AI-powered document understanding.
- Generate interactive Knowledge Graphs.
- Improve information retrieval using Generative AI.
- Secure document management through JWT Authentication.
- Provide enterprise-ready, scalable architecture.
- Enhance engineering productivity through intelligent automation.

5. Key Features

Secure Authentication

JWT-based login, registration, and protected routes.

Document Management

Upload, preview, download, and delete industrial documents.

OCR Processing

Extract text from scanned PDFs and images using Tesseract OCR.

AI Assistant

Ask questions and receive contextual answers powered by Google Gemini AI.

Knowledge Graph

Automatically visualize relationships between extracted entities.

Interactive Graph

Search, highlight nodes, reset view, and export the graph as PNG.

Dashboard Analytics

View document statistics, upload trends, and recent activity.

Theme Support

Light and Dark mode for improved accessibility.

Cloud Storage

Store uploaded files securely using Cloudinary.

Responsive UI

Optimized for desktop and modern web browsers.

6. Technology Stack

Category	Technology	Purpose
Frontend	React (Vite), Tailwind CSS, React Router, React Flow, Axios	User Interface
Backend	Spring Boot, Spring Security, JWT Authentication, REST APIs	REST APIs & Business Logic
Security	Spring Security + JWT	Store Users & Document Metadata Authentication & Authorization
Artificial Intelligence	Google Gemini AI	Intelligent Question Answering
OCR	Tess4J (Tesseract OCR)	Extract Text from PDFs & Images
Database	PostgreSQL (Neon)	Store Users & Document Metadata
Knowledge Graph	Neo4j	Entity Relationship Visualization
Cloud Storage	Cloudinary	Secure PDF Storage
Build Tool	Maven	Dependency Management
Version control	Git & GitHub	Source Code Management
IDE	VS Code	Development Environment

7. System Architecture

IndustrialBrain AI follows a modular three-tier architecture that separates the presentation layer, business logic, and data layer to ensure scalability, maintainability, and secure communication.

Workflow

1. User accesses the React web application.
2. Spring Boot handles authentication and business logic.
3. Documents are uploaded securely to Cloudinary.
4. Tess4J extracts text from uploaded PDF documents.
5. Google Gemini AI analyzes the extracted content and answers user queries.
6. Neo4j generates Knowledge Graph relationships.
7. PostgreSQL (Neon) stores user accounts and document metadata.
8. Dashboard presents analytics and AI-generated insights.

Architecture Diagram

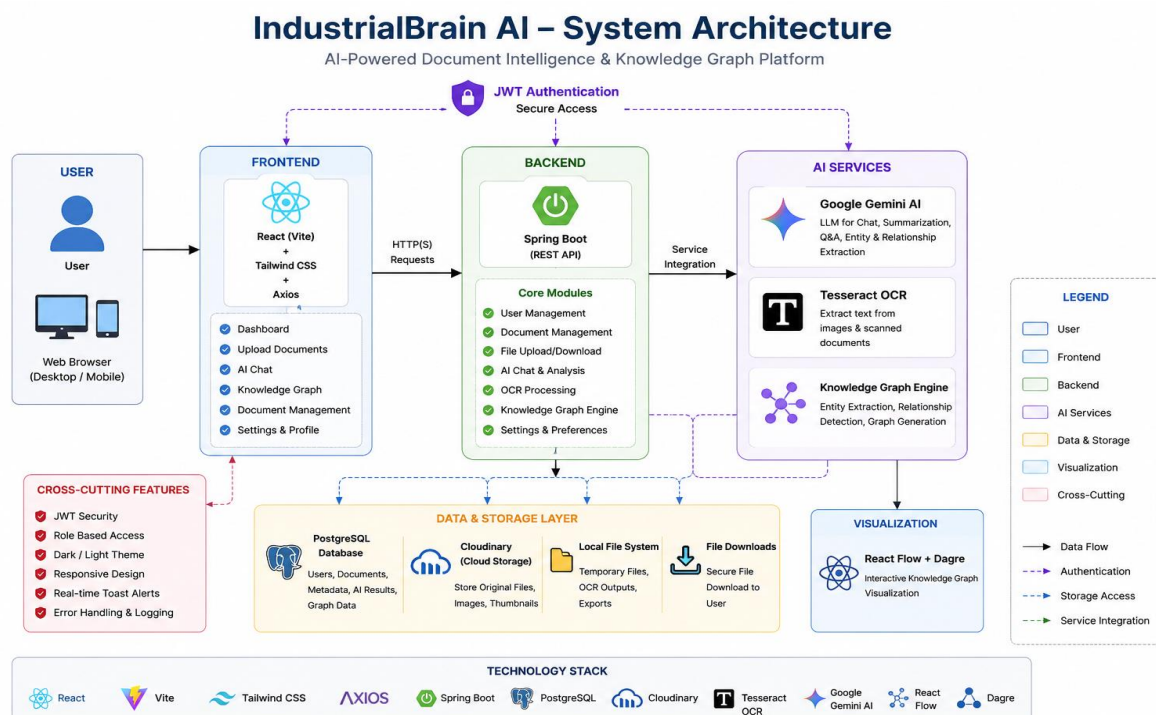


Figure 1: Overall System Architecture showing interactions between the client, backend services, AI modules, cloud storage, and database.

8. Application Workflow

IndustrialBrain AI processes industrial documents through an AI-driven pipeline that transforms static PDF files into searchable knowledge.

Workflow Steps

1. User Login
2. Secure JWT Authentication
3. Upload Industrial PDF
4. Cloudinary File Storage
5. OCR using Tess4J
6. AI Analysis using Google Gemini
7. Knowledge Graph Generation
8. Dashboard Analytics
9. AI Chat for Contextual Question Answering

Workflow Diagram

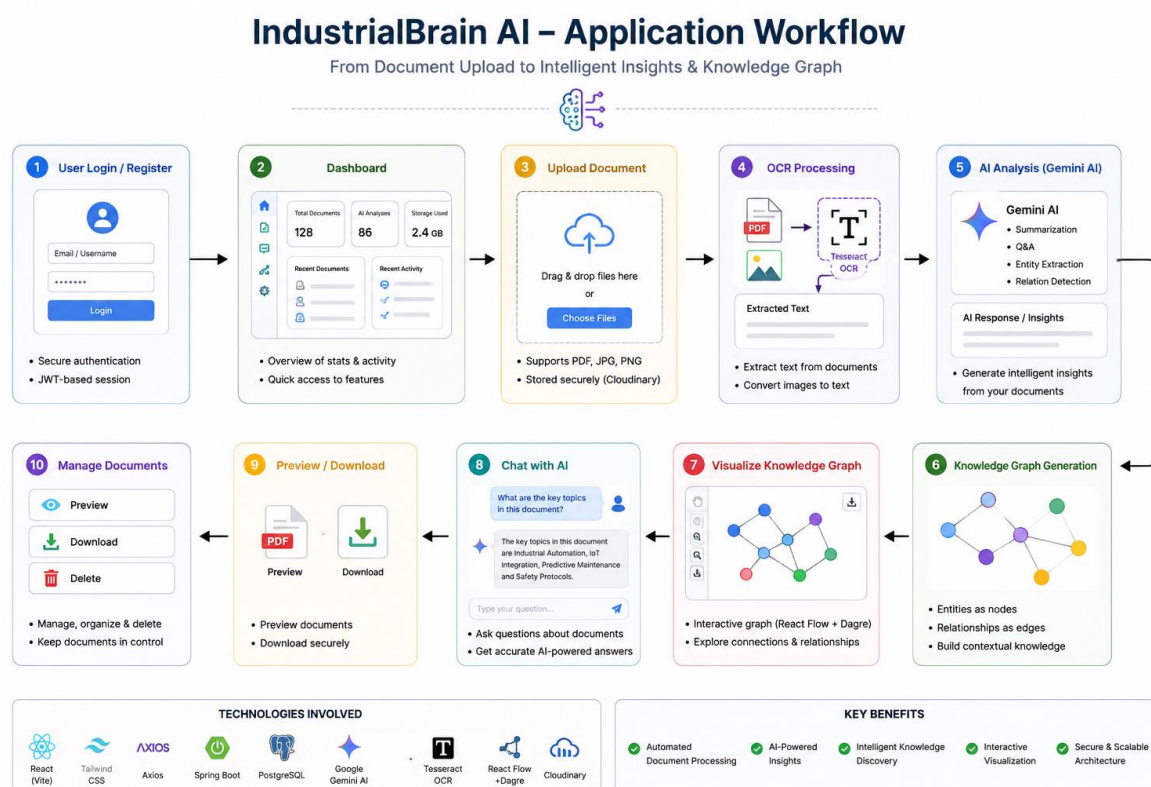


Figure 2: End-to-end workflow from user login to AI analysis and visualization.

9. AI Processing Pipeline

The AI engine processes industrial documents in multiple stages to provide intelligent knowledge retrieval.

Pipeline

- Upload PDF Document
- OCR extracts machine-readable text
- Text preprocessing and cleaning
- Google Gemini analyzes document context
- AI generates contextual responses
- Entity extraction identifies important concepts
- Neo4j builds Knowledge Graph relationships
- Dashboard visualizes insights
- AI Chat answers user questions

Pipeline Diagram

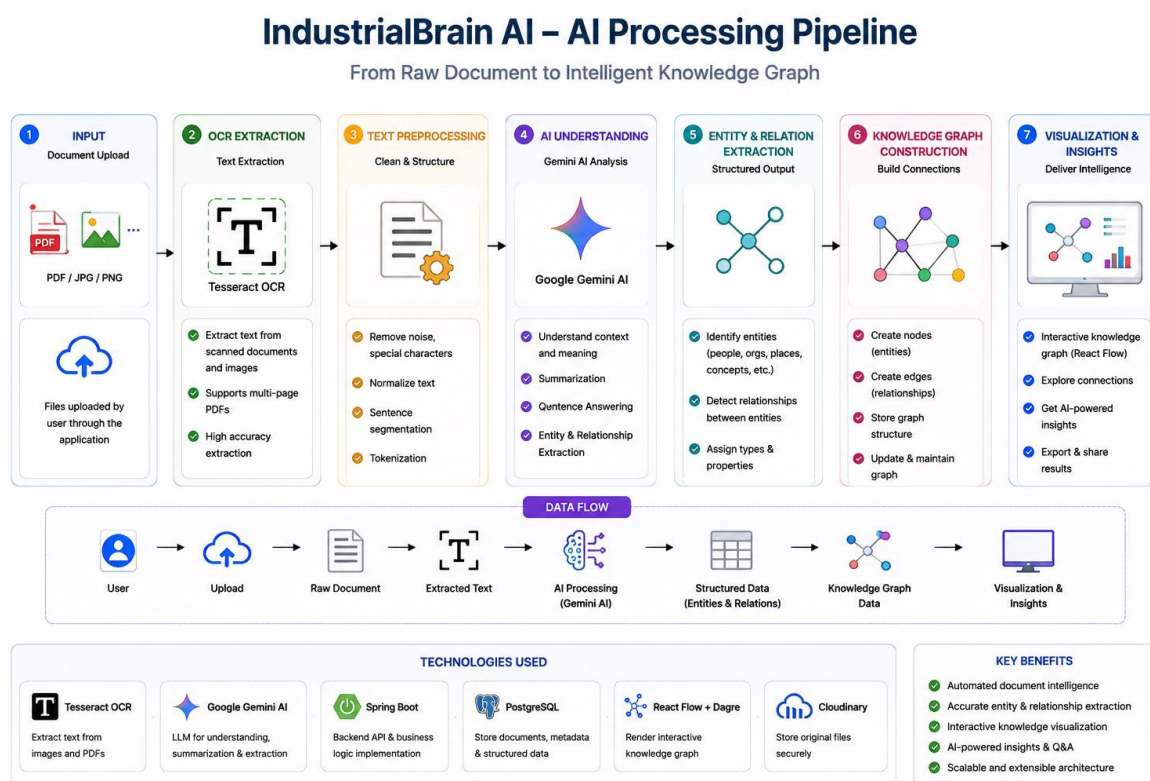


Figure 3: OCR, AI analysis, entity extraction, and knowledge graph generation pipeline.

10. Database Design

IndustrialBrain AI uses a hybrid data storage approach.

PostgreSQL (Neon)

Stores:

- User Accounts
- Authentication Information
- Uploaded Document Metadata
- Activity Logs
- Dashboard Statistics

Neo4j

Stores:

- Document Entities
- Relationships
- Knowledge Graph Structure
- Semantic Connections
- AI-generated Entity Mapping

This hybrid architecture combines relational storage with graph-based knowledge representation to improve search efficiency and contextual understanding.

ER Diagram

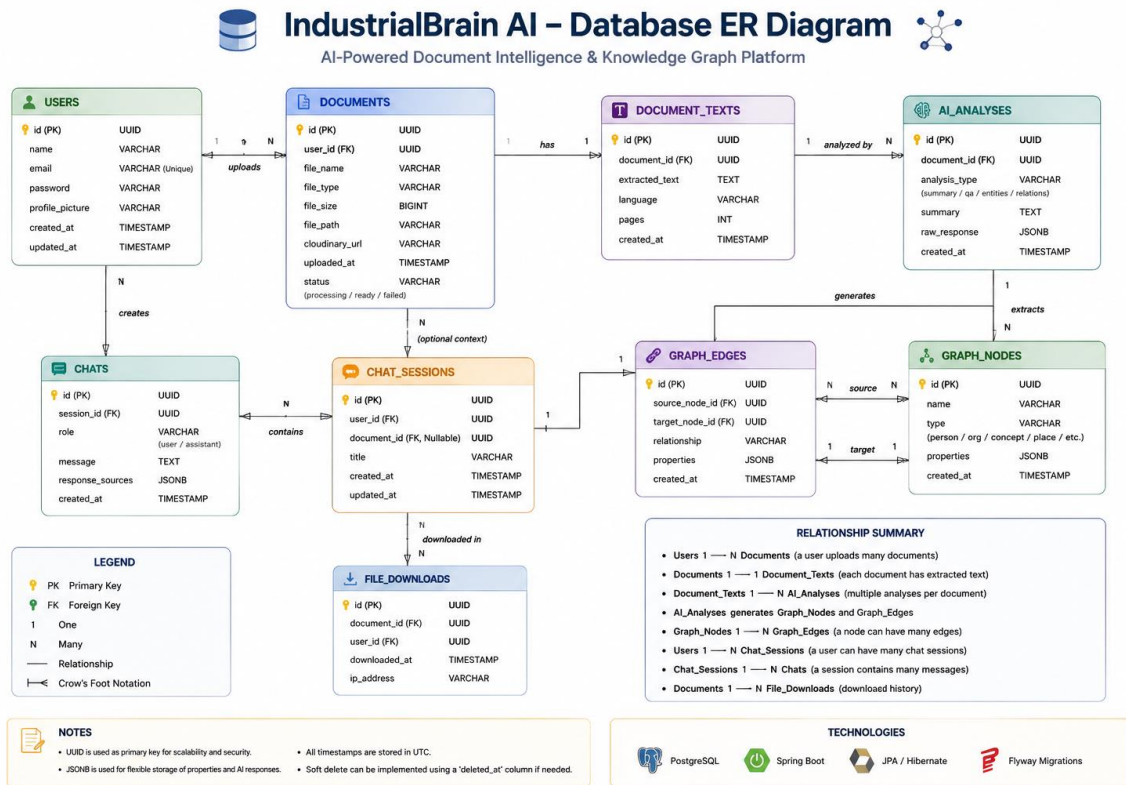


Figure 4: Entity Relationship Diagram illustrating the database schema used by IndustrialBrain AI.

11. Implementation Details

Frontend

The frontend is developed using **React (Vite)** and **Tailwind CSS**, providing a fast, responsive, and modern user interface. It includes secure user authentication, dashboard analytics, document management, AI-powered chat, Knowledge Graph visualization, and responsive layouts for an intuitive user experience. **Axios** is used for API communication, while **React Router** enables seamless navigation between application modules.

Backend

The backend is built using **Spring Boot**, following a modular RESTful architecture. It manages authentication, document processing, OCR integration, AI services, Knowledge Graph generation, and business logic. **Spring Security** with **JWT Authentication** secures all protected APIs, ensuring authorized access to enterprise resources.

Database

IndustrialBrain AI adopts a **hybrid data storage architecture** to efficiently manage structured and interconnected information.

PostgreSQL (Neon)

Stores:

- User accounts
- Authentication details
- Document metadata
- OCR results
- AI-generated responses
- Dashboard statistics

Neo4j

Stores:

- Extracted entities
- Relationships between entities
- Knowledge Graph structure
- Semantic connections
- Graph traversal data

This hybrid approach combines the reliability of relational databases with the flexibility of graph databases for intelligent knowledge discovery.

Cloud Storage

Cloudinary is used for secure cloud storage of uploaded industrial documents. It enables reliable document upload, storage, retrieval, preview, and download while reducing backend storage overhead and improving scalability.

AI Integration

IndustrialBrain AI integrates multiple AI technologies to transform static documents into intelligent knowledge.

Google Gemini AI

- Context-aware document understanding
- Natural language question answering
- AI-powered document explanations
- Intelligent content summarization
- Contextual response generation

Tess4J OCR

- Text extraction from scanned PDFs
- OCR processing for image-based documents
- Machine-readable text generation
- Preprocessing for AI analysis

System Integration

The complete processing workflow follows these steps:

1. User logs in using JWT Authentication.
2. Industrial PDF documents are uploaded securely.
3. Cloudinary stores uploaded documents.
4. Tess4J extracts text from scanned documents.
5. Google Gemini AI analyzes extracted content.
6. Entities are extracted and stored in Neo4j.
7. PostgreSQL (Neon) stores document metadata and AI results.
8. Dashboard visualizes analytics and document insights.
9. AI Chat provides contextual answers based on processed knowledge.

12. Project Screenshots

The following screenshots demonstrate the key functionality implemented in IndustrialBrain AI.

Authentication

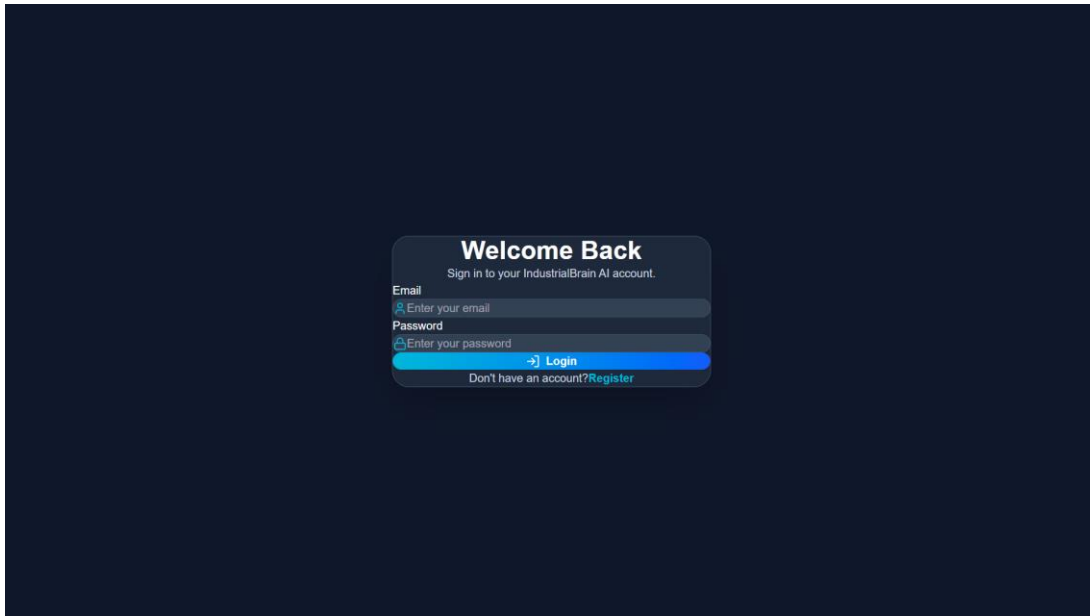


Figure 12.1: Secure User Login with JWT Authentication

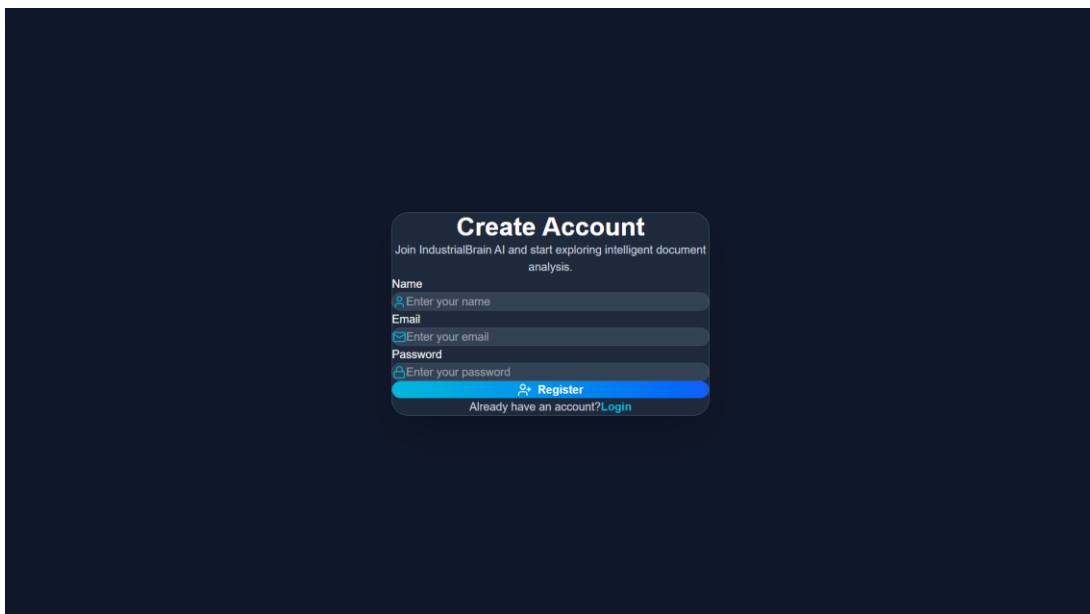


Figure 12.2: New User Registration Interface

Dashboard

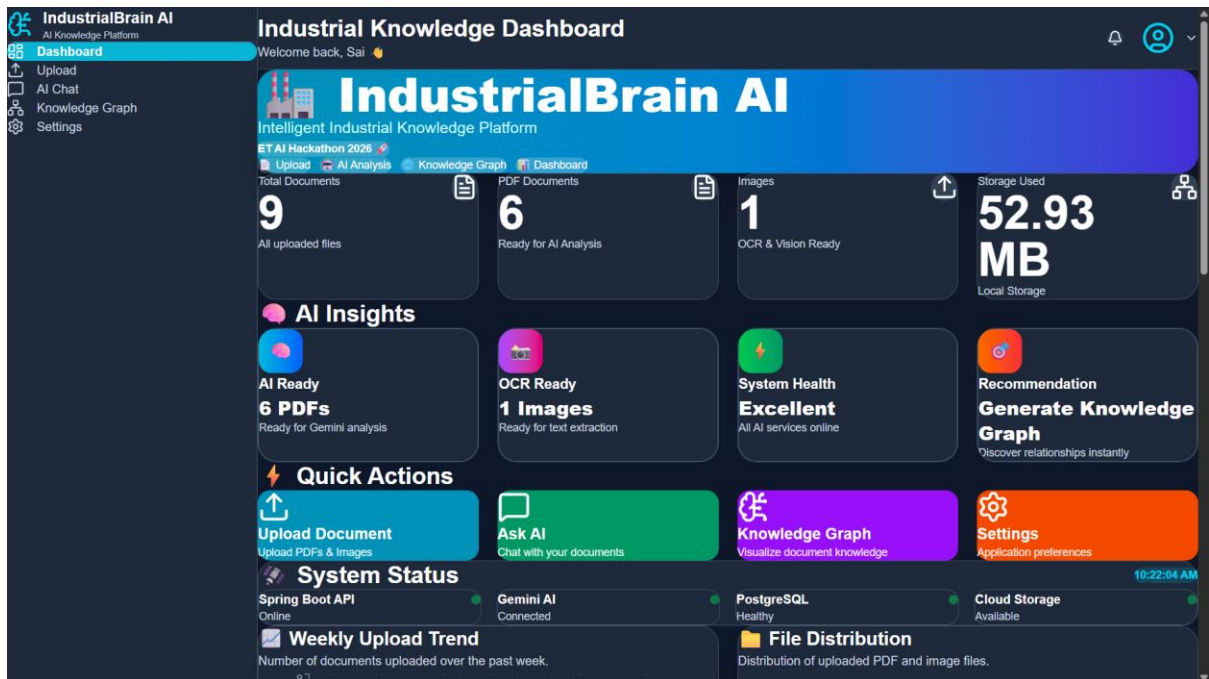


Figure 12.3: IndustrialBrain AI Dashboard with Document Analytics



Figure 12.4: Dashboard - Search

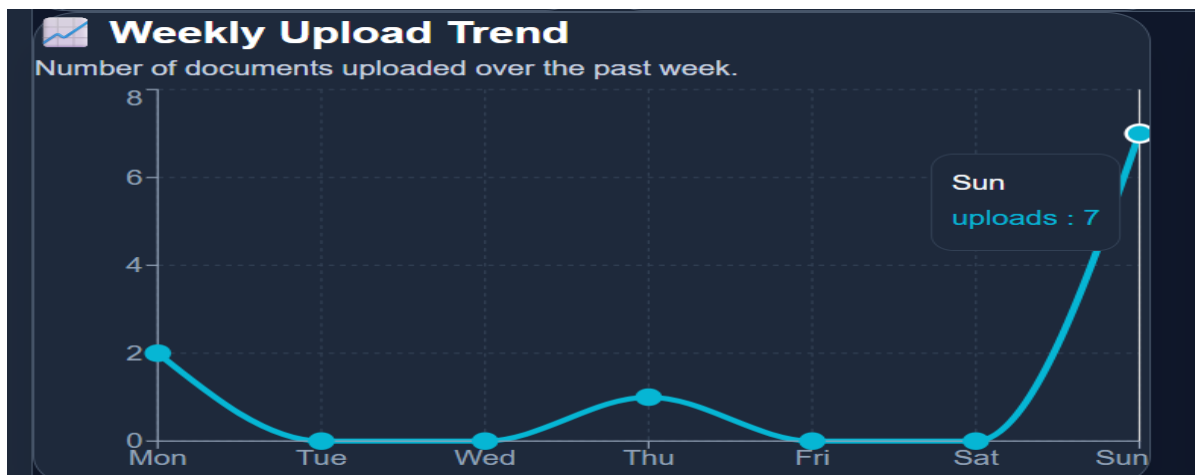


Figure 12.5: Weekly Upload Trend

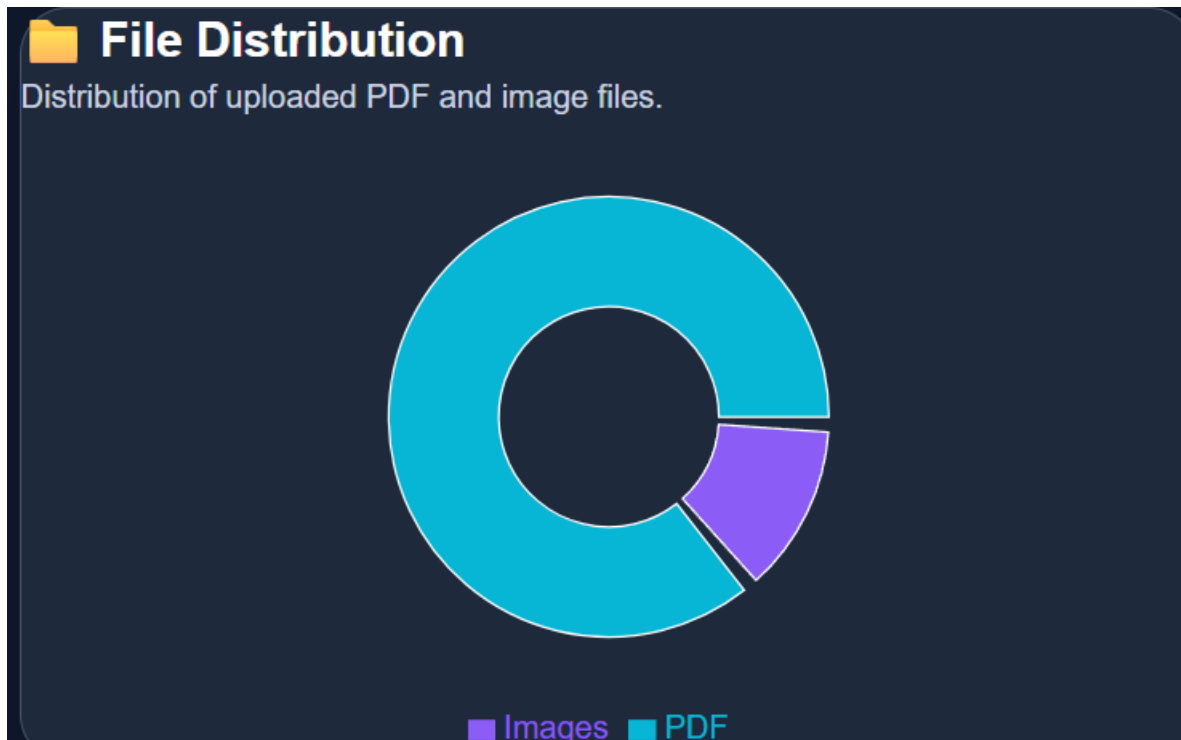


Figure 12.6: File Distribution

Recent Activity	
Latest document uploads and actions	
	ET AI HACKATHON 2026.pdf Document Uploaded 7/20/2026, 10:27:06 AM
	OCR_Scanned_Document_DPI_120.pdf Document Uploaded 7/20/2026, 5:27:28 AM
	Ananta.width-1000.format-webp.jpg Document Uploaded 7/19/2026, 10:45:10 PM
	IndustrialBrain-AI-Project-Documentation.docx Document Uploaded 7/19/2026, 9:53:54 PM
	Trees.pdf Document Uploaded 7/19/2026, 8:27:59 PM

Figure 12.7: Recent Activity

Document Upload:

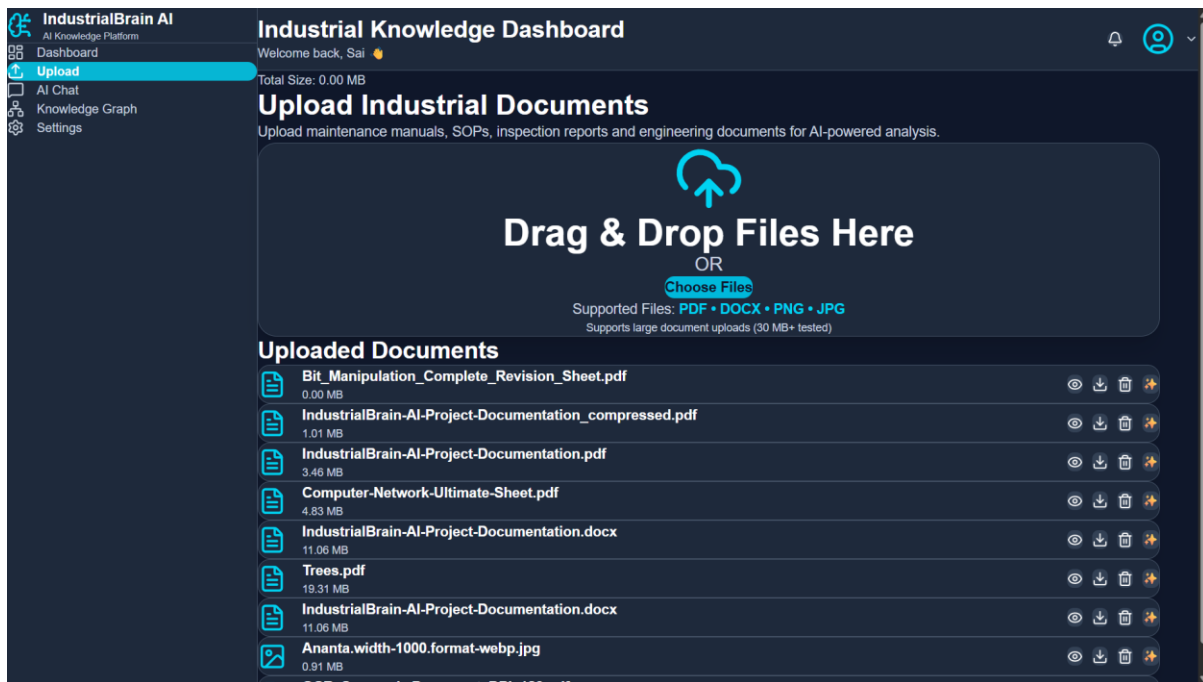


Figure 12.8: Upload Industrial PDF Documents

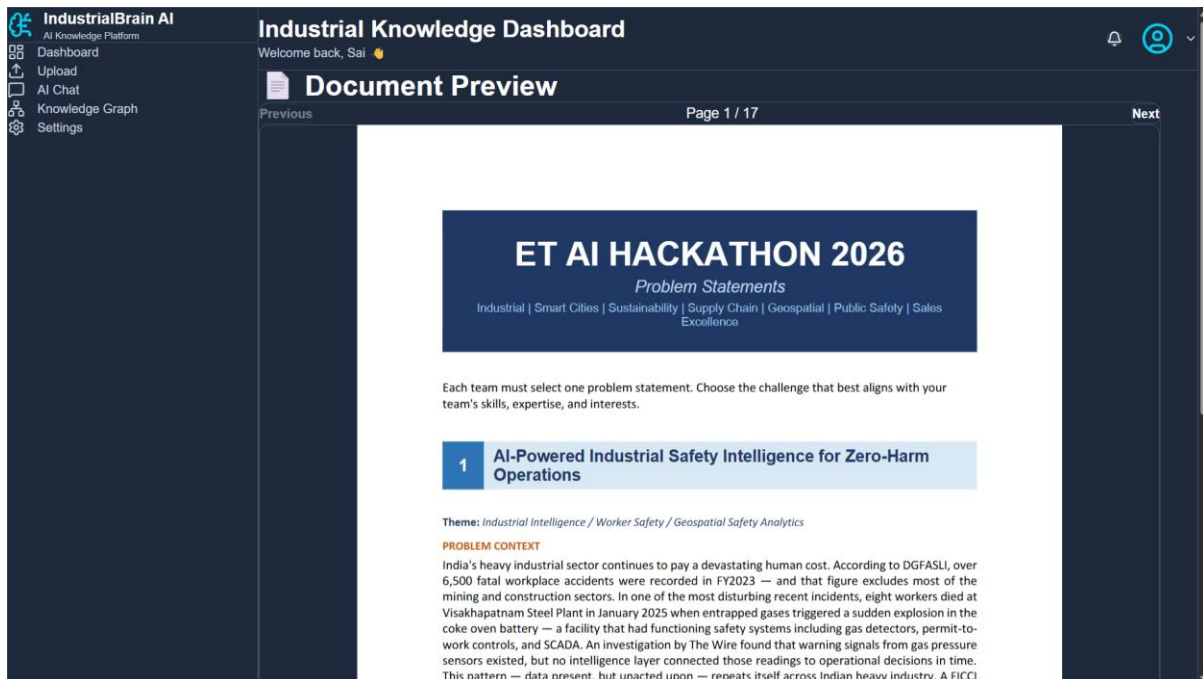


Figure 12.9: PDF Preview Before AI Processing

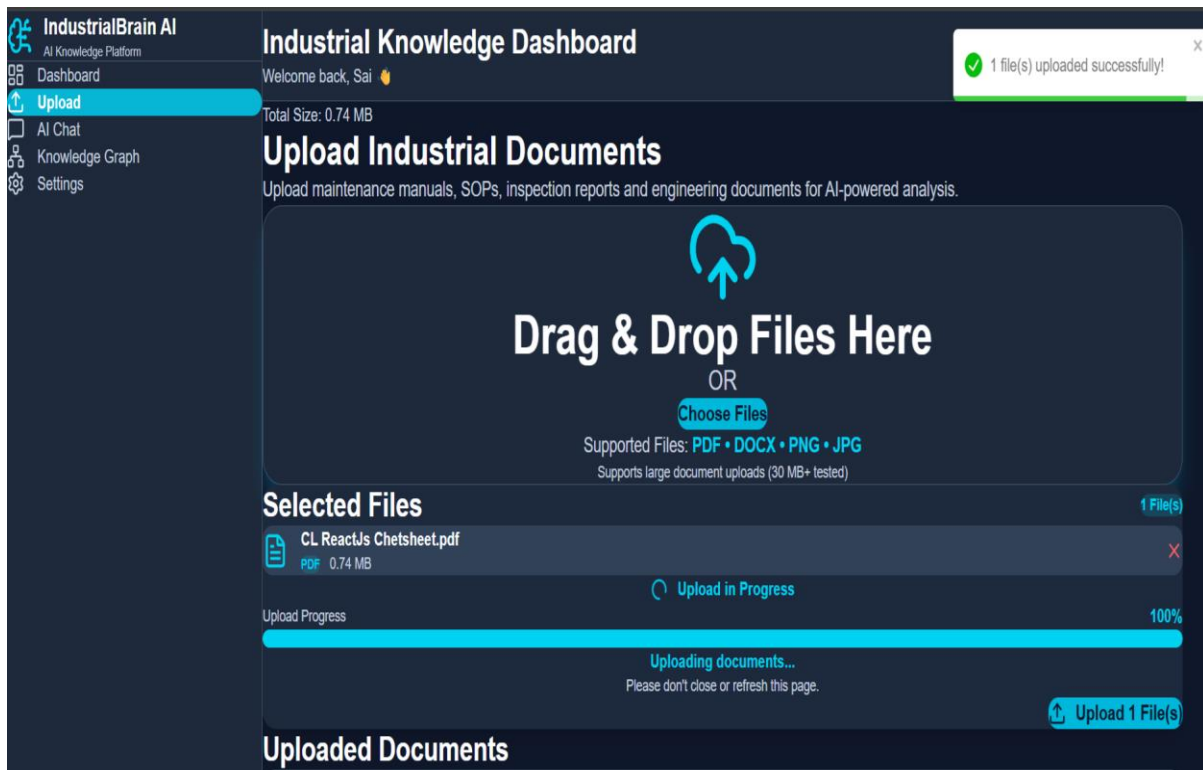


Figure 12.10: Upload Success



Figure 12.11: Document List

IndustrialBrain AI

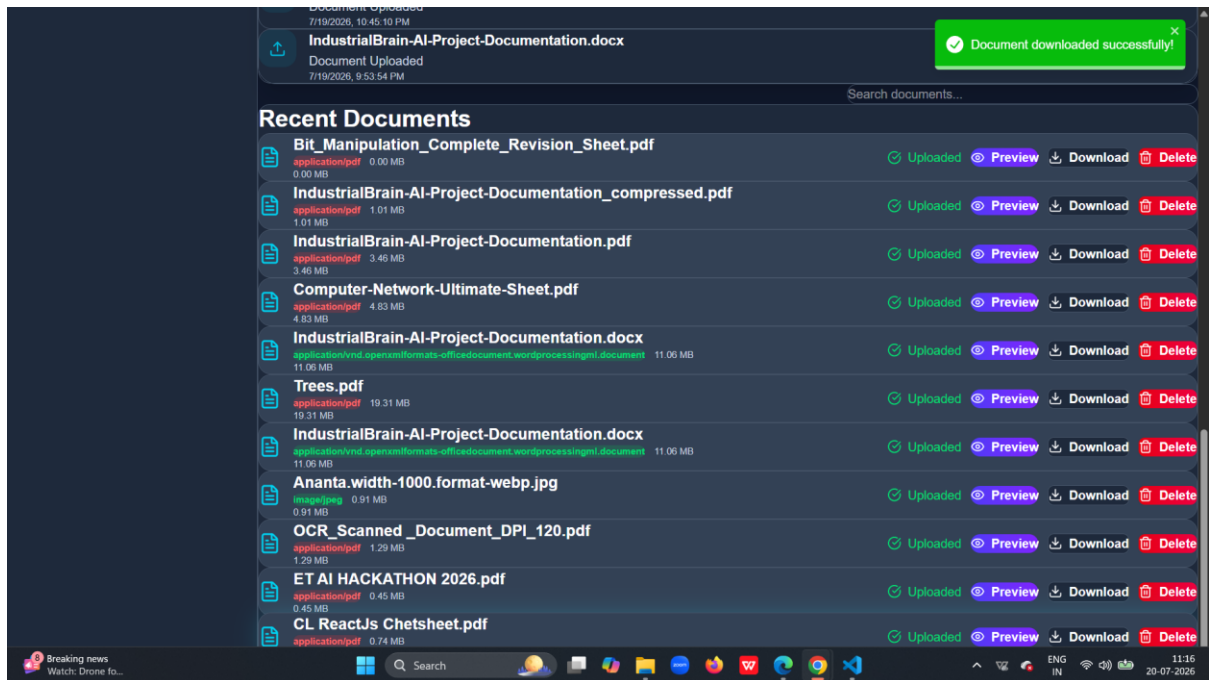


Figure 12.12: Document Download Success

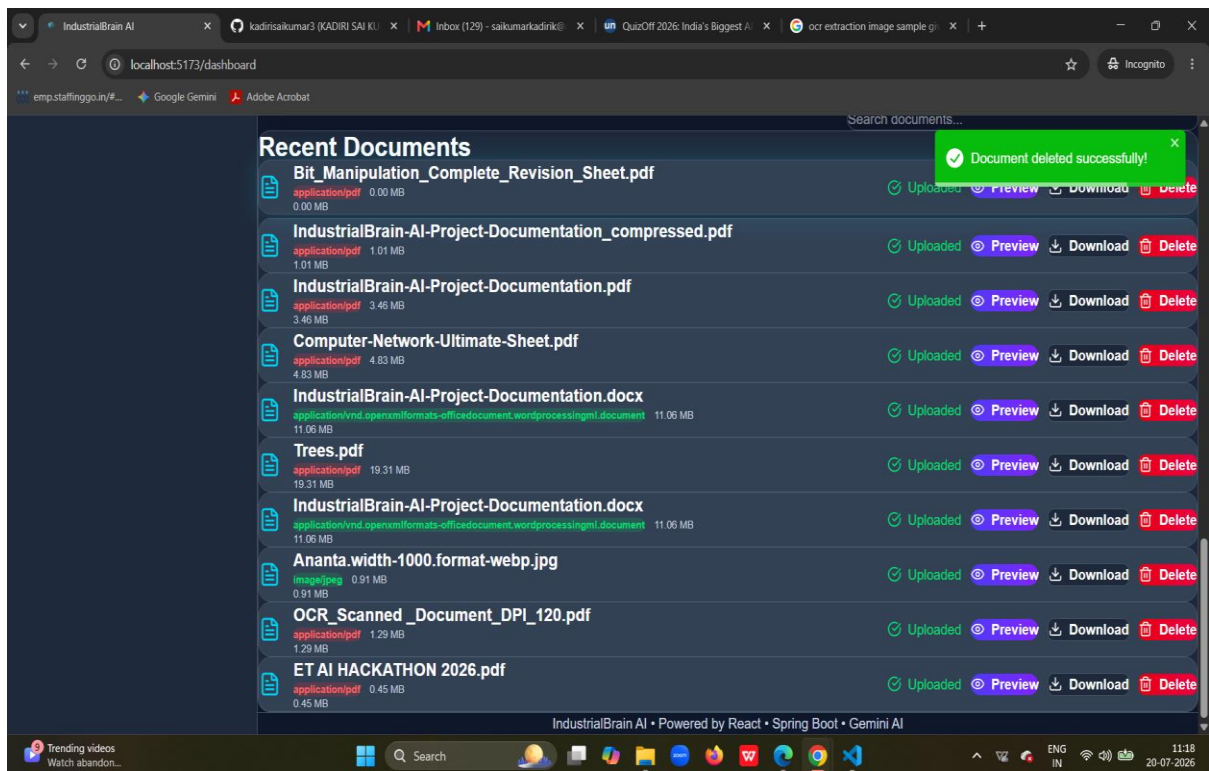


Figure 12.13: Document Deleted Successfully

Artificial Intelligence

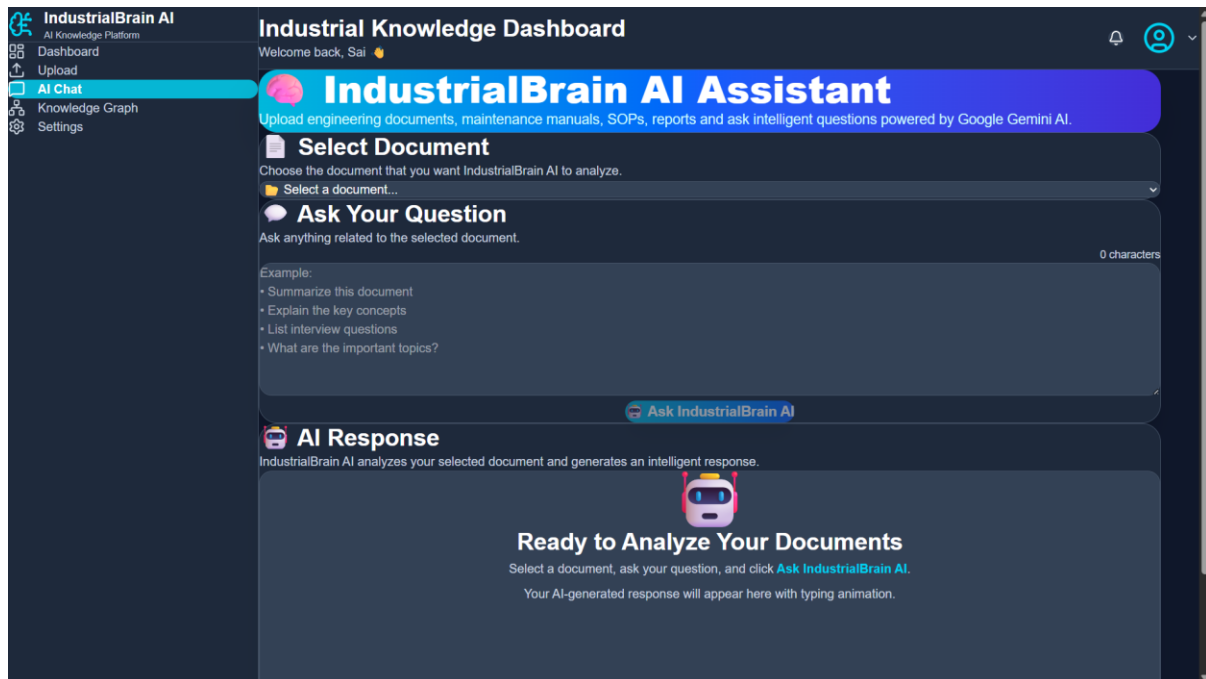


Figure 12.14: AI Assistant for Context-Aware Question Answering

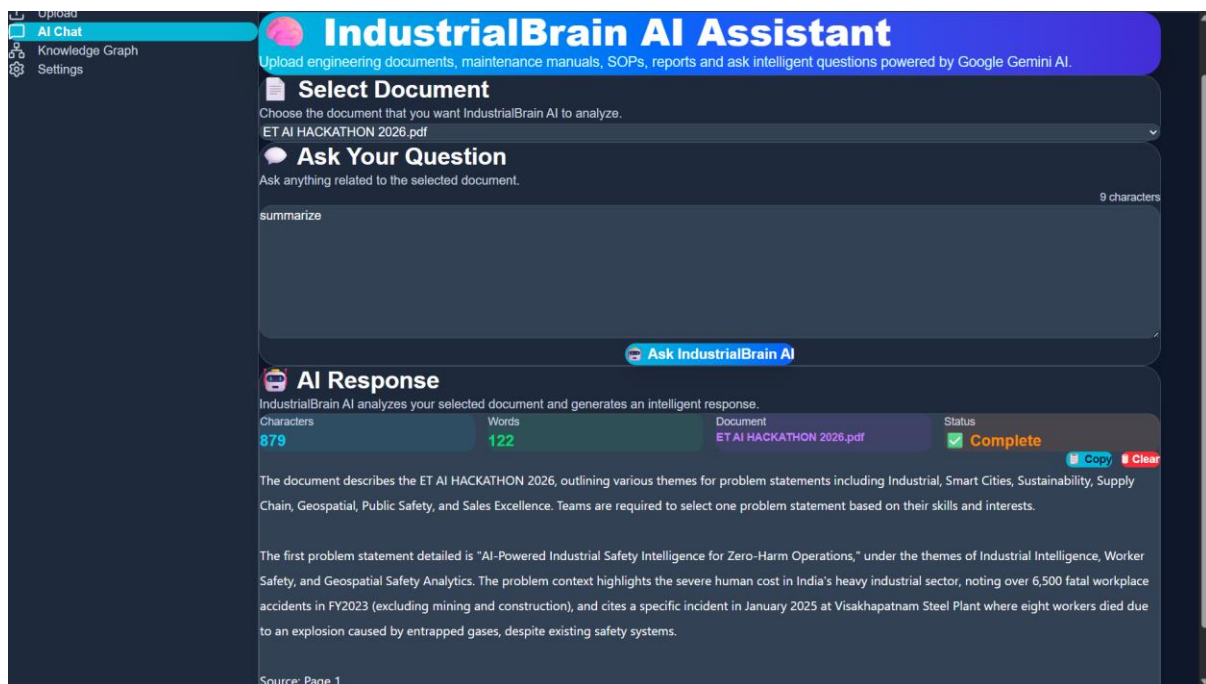


Figure 12.15: Google Gemini AI Generated Response

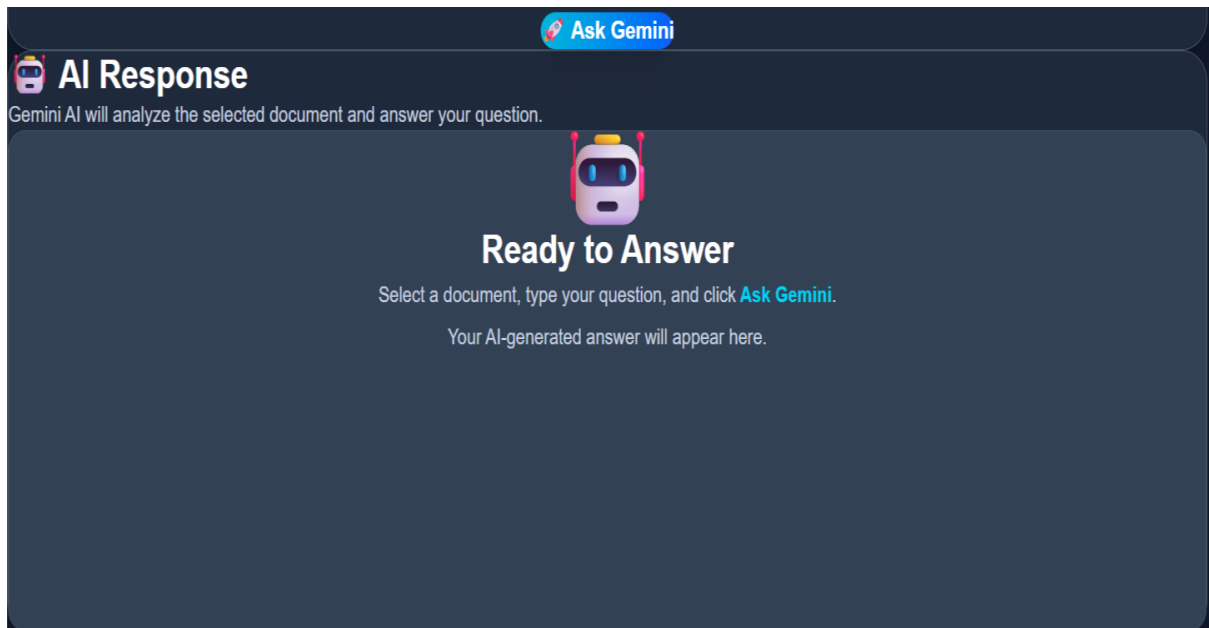


Figure 12.16: Gemini AI Response

Knowledge Graph

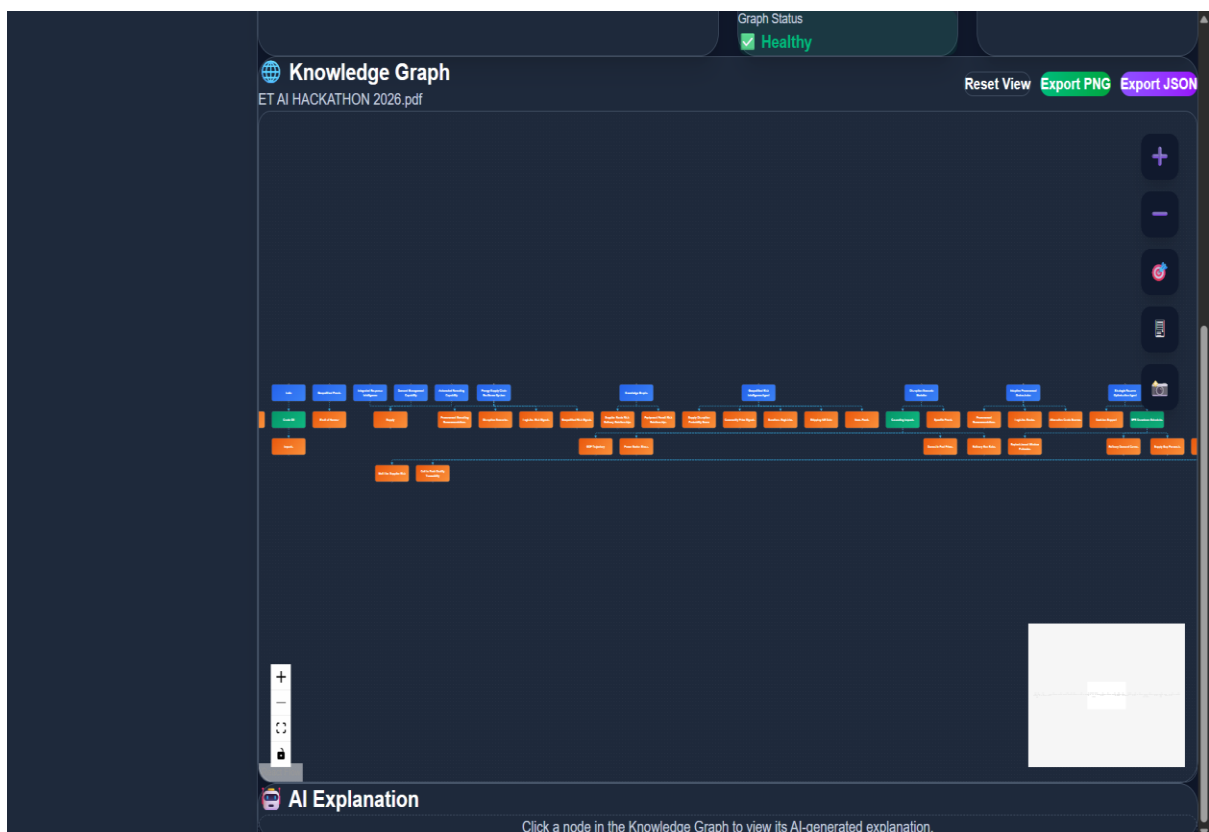


Figure 12.17: Interactive Knowledge Graph Visualization

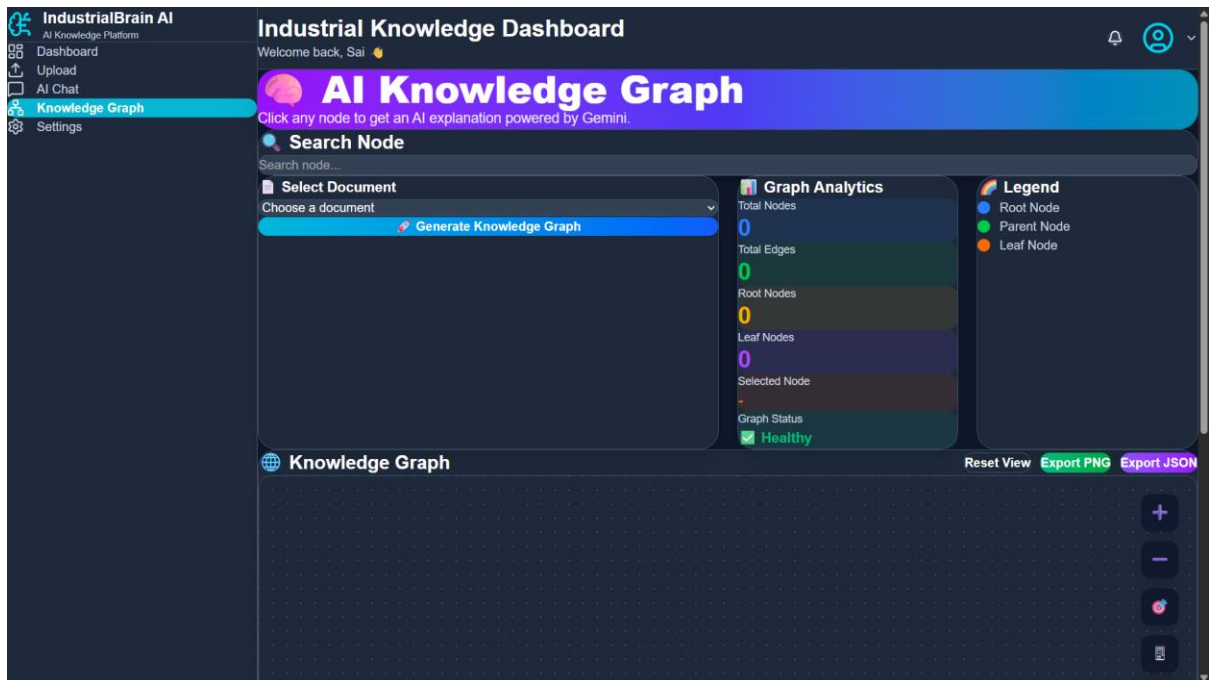
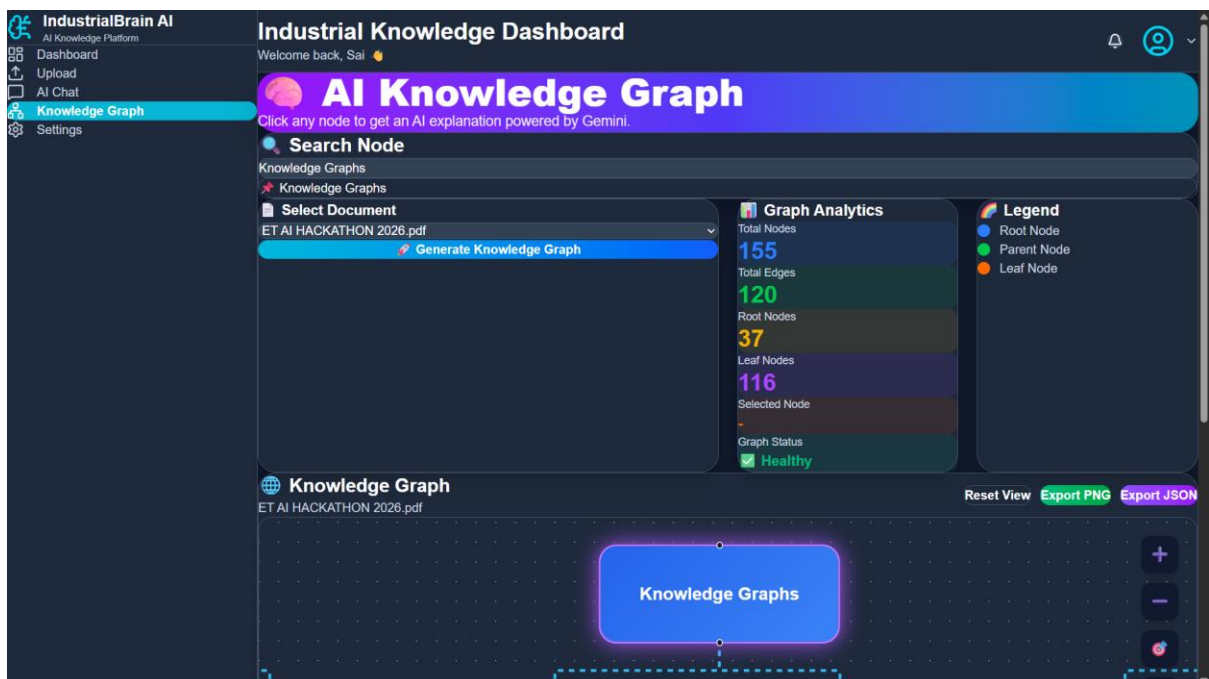


Figure 12.18: Knowledge Overview



12.19: Node Search

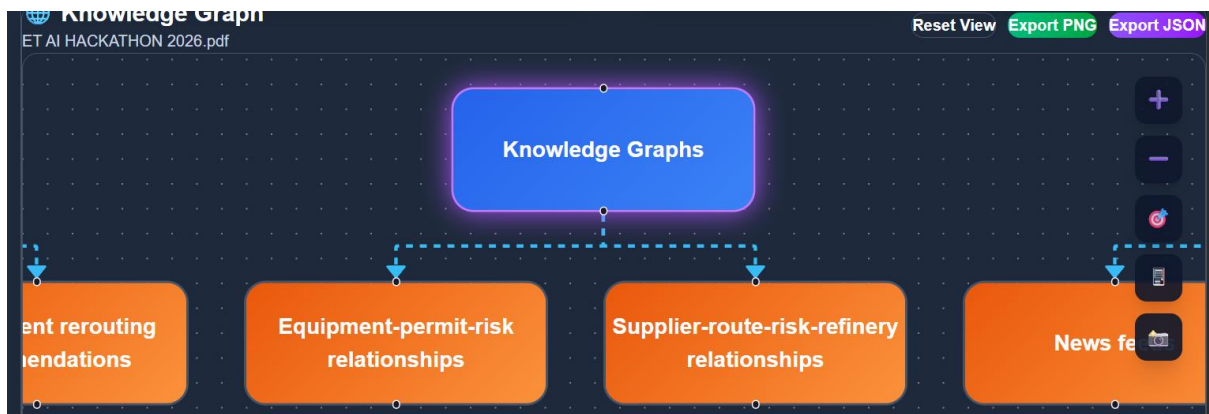


Figure 12.20: Node Highlight

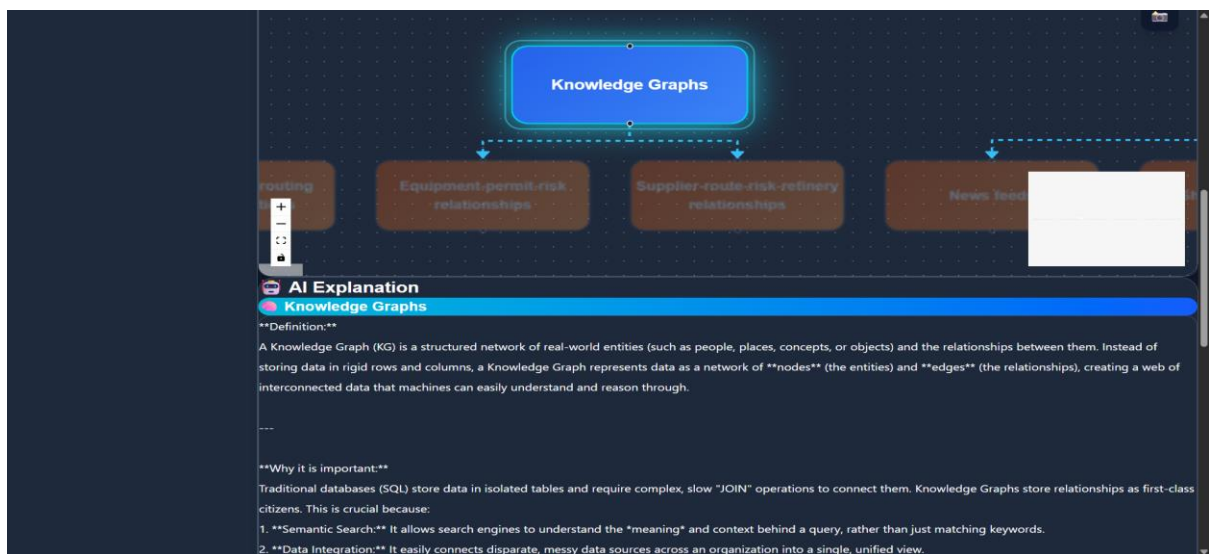


Figure 12.21: Node AI Explanation

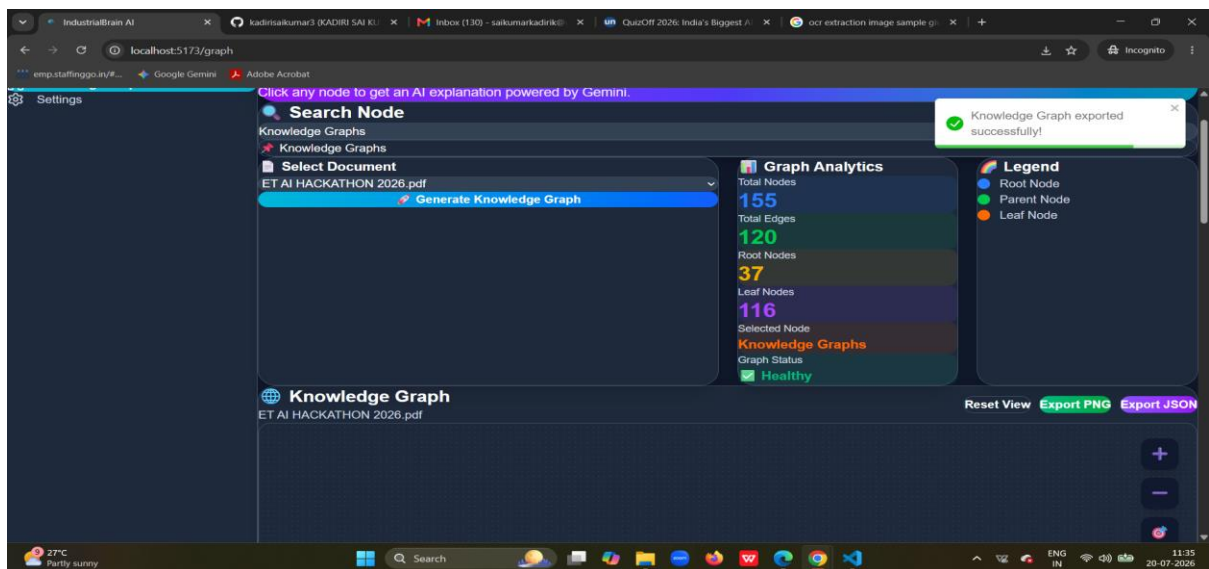


Figure 12.22: Export PNG Successful

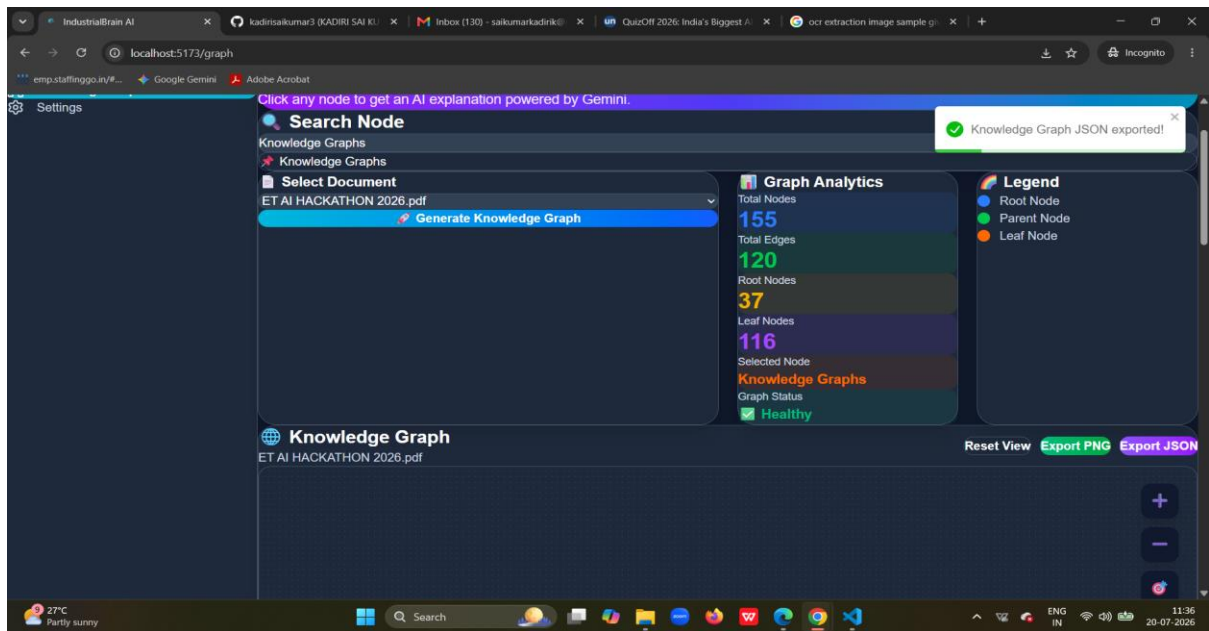


Figure 12.23: Export JSON Successful

Profile & Settings

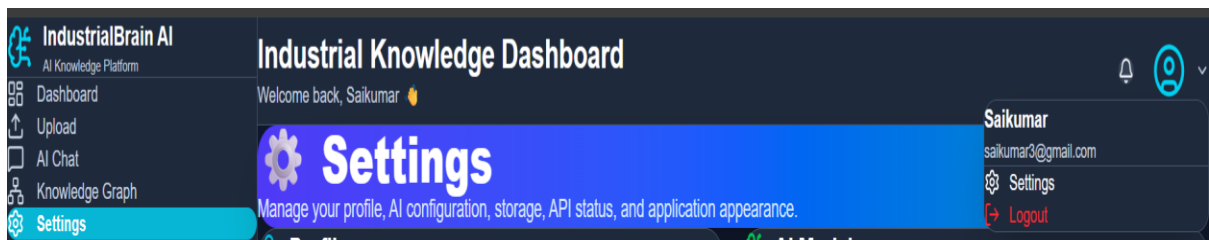


Figure 12.24: Profile Menu

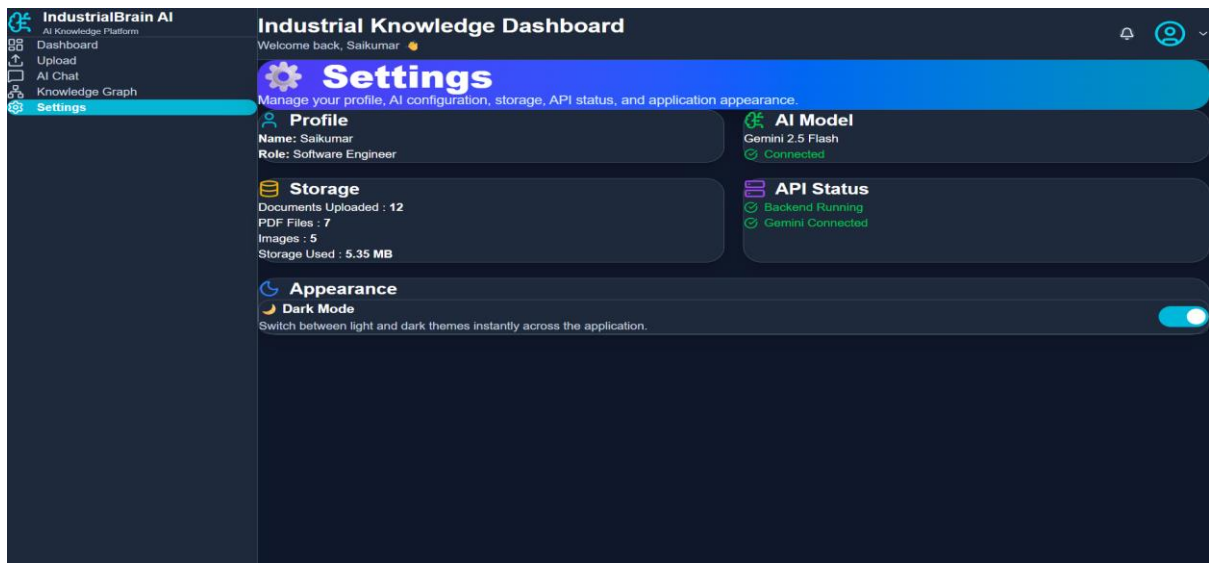


Figure 12.25: Settings Page

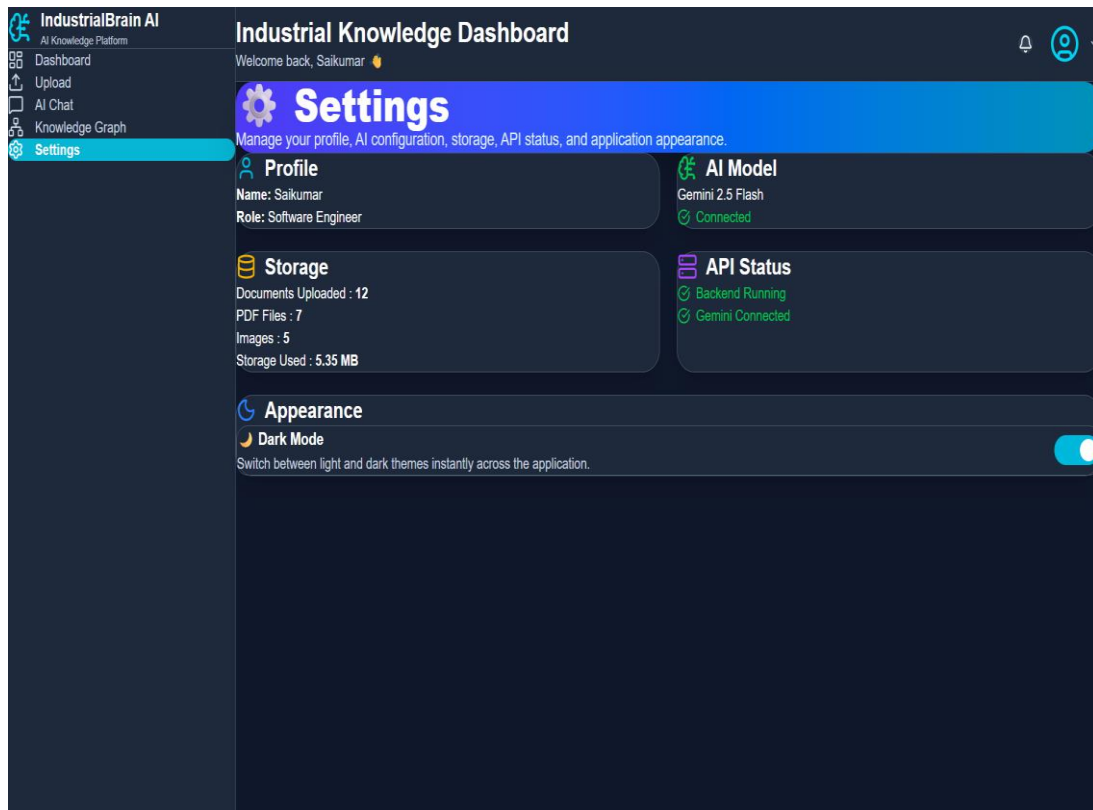


Figure 12.26: Dark Theme

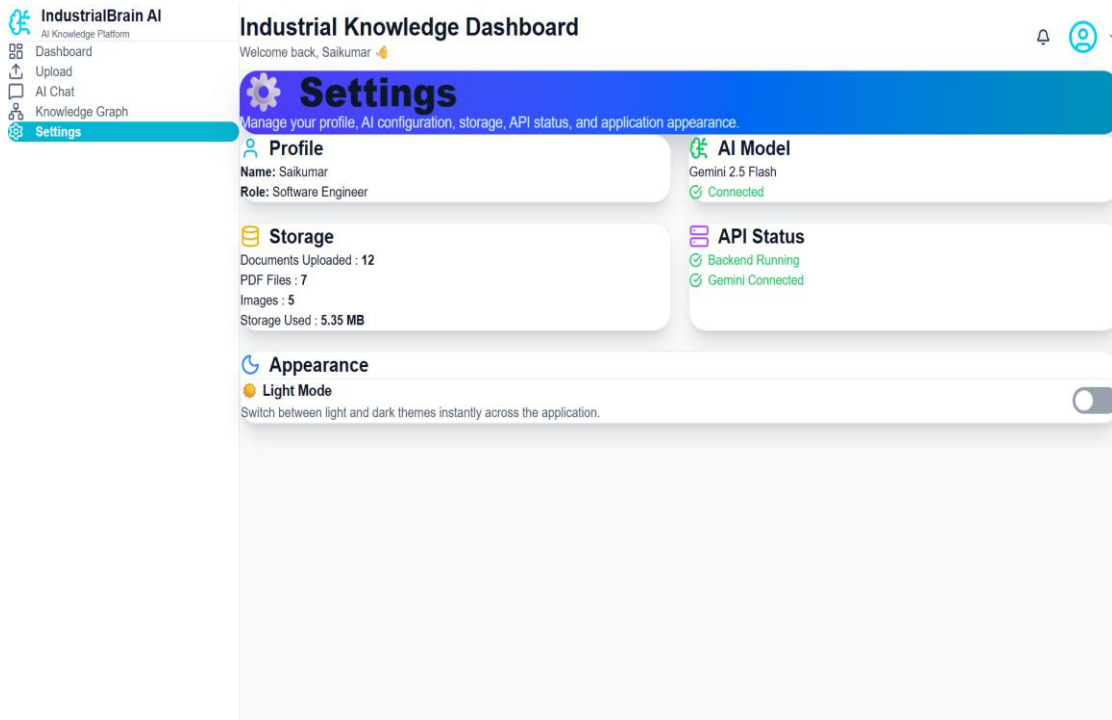


Figure 12.27: Light Theme

14. Security Features

IndustrialBrain AI follows enterprise-grade security practices to protect user data, documents, and AI services.

- JWT-based Authentication and Authorization.
- Spring Security for secure REST API protection.
- Password encryption using BCrypt hashing.
- Role-based access to authenticated resources.
- Secure document storage using Cloudinary.
- Input validation and global exception handling.
- CORS configuration for secure frontend-backend communication.
- HTTPS-ready architecture for secure deployment.
- Secure AI service integration with protected API keys.
- Session-independent authentication using JWT tokens.

15. Challenges Faced and Solutions

Challenge	Solution
OCR accuracy on scanned and low-quality documents	Implemented Tess4J OCR with image preprocessing techniques
Visualizing relationships between extracted entities	Developed an interactive Knowledge Graph using Neo4j
Managing structured and graph-based data	Adopted a hybrid architecture using PostgreSQL (Neon) and Neo4j
Building a scalable application	Designed a modular React + Spring Boot client-server architecture
Secure REST API communication	Implemented Spring Security with JWT-based authorization
Understanding technical industrial documents	Integrated Google Gemini AI for contextual analysis

16. Future Enhancements

IndustrialBrain AI has been designed with extensibility in mind. The current implementation establishes a strong foundation for intelligent industrial knowledge management while allowing future capabilities to be integrated with minimal architectural changes.

Future enhancements include:

- Support multilingual OCR and AI-powered document analysis.
- Enable voice-based interaction with industrial documents.
- Provide real-time collaboration for multiple users.
- Integrate predictive maintenance recommendations using AI and Machine Learning.
- Support additional document formats, including CAD drawings, Excel spreadsheets, and engineering diagrams.
- Introduce advanced analytics dashboards with KPI reporting.
- Implement enterprise-grade Role-Based Access Control (RBAC).
- Deploy using Docker and Kubernetes for scalable cloud infrastructure.
- Integrate IoT sensor data with the Knowledge Graph for contextual industrial insights.
- Develop mobile applications for Android and iOS field engineers.
- Introduce AI-powered document summarization and automatic report generation.

17. Conclusion

IndustrialBrain AI demonstrates how Artificial Intelligence can transform traditional industrial document management into an intelligent knowledge platform. By combining OCR, Google Gemini AI, Knowledge Graph visualization, secure cloud storage, and modern web technologies, the platform enables users to efficiently discover, understand, and utilize industrial knowledge.

The application integrates React, Spring Boot, PostgreSQL (Neon), Neo4j, Tess4J OCR, Cloudinary, and JWT Authentication into a scalable, secure, and enterprise-ready architecture. Its modular design supports future enhancements while maintaining high performance, maintainability, and usability.

IndustrialBrain AI successfully addresses the ET AI Hackathon challenge by applying AI to solve real-world industrial problems. The platform reduces manual effort, improves knowledge accessibility, accelerates decision-making, and provides a practical foundation for intelligent digital transformation in industrial environments.

18. References

- [Official Google Gemini AI Documentation](#)
- [Official Tess4J Documentation](#)
- [Official Tesseract OCR Documentation](#)
- [Official React Documentation](#)
- [Official Spring Boot Documentation](#)
- [Official Spring Security Documentation](#)
- [Official Tailwind CSS Documentation](#)
- [Official PostgreSQL \(Neon\) Documentation](#)
- [Official Neo4j Documentation](#)
- [Official Cloudinary Documentation](#)
- [Official JWT \(JSON Web Token\) Documentation](#)
- [Official Maven Documentation](#)
- [Official Git Documentation](#)
- [Official GitHub Documentation](#)
- [Official React Flow Documentation](#)

19. Repository Structure

IndustrialBrain-AI

```
|
|├── frontend/
|   |├── src/
|   |├── public/
|   └── package.json
|
|├── spring-backend/
|   |├── src/
|   |├── pom.xml
|   └── uploads/
|
|├── docs/
|
|├── screenshots/
|
|├── architecture/
|
|├── README.md
|
└── LICENSE
```

20. Acknowledgements

The successful development of **IndustrialBrain AI** was made possible through the support of modern open-source technologies, official documentation, and the **ET AI Hackathon 2026**, which inspired innovation in solving real-world industrial knowledge management challenges.

Special thanks to the developers and maintainers of **React**, **Spring Boot**, **Google Gemini AI**, **Tess4J**, **Neo4j**, **PostgreSQL (Neon)**, **Cloudinary**, and the open-source community for providing the tools and resources that enabled the successful implementation of this project.

Presented & Developed By

Sai Kumar Kadiri

Software Engineer

Project: IndustrialBrain AI

Event: ET AI Hackathon 2026 – AI Innovation Challenge